

**KRASNOSHCHELYE, Russian Federation**

WMO: 222350

Lat: **67.3495N**Lon: **37.0529E**Elev: **516**StdP: **14.42**Time Zone: **3.00 (E03)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 2             | -28.1      | -21.7 | -32.8                         | 0.9 | -28.1 | -26.1 | 1.3 | -21.7 | 18.2                  | 27.6 | 16.2 | 25.3 | 1.5                      | 290  | 0.721 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 16.5                   | 76.8            | 62.4 | 72.5 | 60.0 | 68.7 | 58.2 | 64.2                | 72.9 | 61.9 | 70.1 | 59.3 | 66.6 | 6.3                  | 220  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 60.8                            | 81.0 | 67.9 | 58.1 | 73.6 | 65.6 | 55.7 | 67.3 | 63.1 | 29.7          | 72.8 | 28.0 | 70.1 | 26.2 | 66.7 | 72.7              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 16.1          | 13.9 | 12.0 | DB                         | -33.9 | 81.0               | 6.8 | 3.8                                                | -38.8 | 83.7       | -42.8 | 86.0       | -46.7 | 88.1       | -51.6 | 90.9 | (4) |
| (5)               |      |      | WB                         | -34.1 | 66.8               | 6.8 | 3.1                                                | -39.0 | 69.1       | -42.9 | 70.9       | -46.8 | 72.6       | -51.7 | 74.8 | (5) |

**Monthly Climatic Design Conditions**

|      |                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct  | Nov   | Dec   |      |
|------|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|-------|-------|------|
|      |                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)  | (o)   | (p)   |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 31.0   | 8.9   | 8.3   | 16.6  | 27.6  | 38.3  | 49.3  | 56.5  | 52.1  | 43.9  | 32.8 | 21.2  | 15.3  | (6)  |
| (7)  |                                                                                  | DBStd       | 19.27  | 14.69 | 15.05 | 10.90 | 8.67  | 8.25  | 7.94  | 7.35  | 6.59  | 6.25  | 7.64 | 10.91 | 13.02 | (7)  |
| (8)  |                                                                                  | HDD50       | 7376   | 1275  | 1168  | 1034  | 671   | 375   | 108   | 22    | 50    | 201   | 534  | 864   | 1074  | (8)  |
| (9)  |                                                                                  | HDD65       | 12415  | 1740  | 1588  | 1499  | 1121  | 829   | 473   | 278   | 403   | 632   | 999  | 1314  | 1539  | (9)  |
| (10) |                                                                                  | CDD50       | 459    | 0     | 0     | 0     | 0     | 11    | 89    | 224   | 116   | 19    | 0    | 0     | 0     | (10) |
| (11) |                                                                                  | CDD65       | 23     | 0     | 0     | 0     | 0     | 0     | 3     | 15    | 5     | 0     | 0    | 0     | 0     | (11) |
| (12) |                                                                                  | CDH74       | 242    | 0     | 0     | 0     | 0     | 3     | 44    | 146   | 49    | 0     | 0    | 0     | 0     | (12) |
| (13) |                                                                                  | CDH80       | 32     | 0     | 0     | 0     | 0     | 0     | 5     | 20    | 7     | 0     | 0    | 0     | 0     | (13) |
| (14) | Wind                                                                             | WSAvg       | 5.5    | 5.2   | 5.3   | 5.9   | 6.0   | 6.0   | 6.0   | 5.2   | 5.0   | 5.4   | 5.7  | 5.3   | 5.5   | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 20.4   | 1.2   | 0.9   | 1.1   | 1.1   | 1.6   | 2.1   | 2.5   | 2.8   | 2.0   | 2.0  | 1.7   | 1.3   | (15) |
| (16) |                                                                                  | PrecMax     | 30.7   | 2.6   | 1.9   | 2.2   | 3.0   | 3.1   | 4.3   | 5.4   | 5.9   | 4.0   | 4.2  | 3.5   | 3.1   | (16) |
| (17) |                                                                                  | PrecMin     | 13.4   | 0.4   | 0.3   | 0.3   | 0.1   | 0.7   | 0.3   | 0.0   | 0.9   | 0.0   | 0.4  | 0.6   | 0.6   | (17) |
| (18) |                                                                                  | PrecStd     | 3.9    | 0.6   | 0.4   | 0.5   | 0.6   | 0.6   | 0.9   | 1.3   | 1.2   | 0.9   | 1.0  | 0.7   | 0.6   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 34.5  | 35.5  | 40.4  | 50.8  | 70.1  | 79.5  | 82.9  | 79.5  | 65.4  | 50.7 | 41.5  | 36.1  | (19) |
| (20) |                                                                                  |             | MCWB   | 33.2  | 33.4  | 35.1  | 42.9  | 55.0  | 61.3  | 65.5  | 64.5  | 56.6  | 48.4 | 40.9  | 34.3  | (20) |
| (21) |                                                                                  | 2%          | DB     | 32.0  | 32.6  | 35.4  | 45.4  | 62.4  | 73.1  | 78.3  | 72.9  | 61.5  | 47.5 | 37.6  | 34.1  | (21) |
| (22) |                                                                                  |             | MCWB   | 31.4  | 31.5  | 32.7  | 39.0  | 50.7  | 58.5  | 63.4  | 60.9  | 54.6  | 45.8 | 36.9  | 33.1  | (22) |
| (23) |                                                                                  | 5%          | DB     | 29.4  | 30.4  | 33.0  | 41.7  | 57.0  | 68.6  | 74.3  | 68.0  | 57.2  | 45.2 | 35.2  | 32.5  | (23) |
| (24) |                                                                                  |             | MCWB   | 28.7  | 29.4  | 31.0  | 36.7  | 47.8  | 56.1  | 61.7  | 59.1  | 52.3  | 43.7 | 34.5  | 31.7  | (24) |
| (25) |                                                                                  | 10%         | DB     | 26.9  | 27.5  | 31.2  | 39.0  | 52.3  | 64.2  | 70.7  | 63.5  | 54.0  | 43.0 | 33.5  | 30.6  | (25) |
| (26) |                                                                                  |             | MCWB   | 26.2  | 26.5  | 29.4  | 34.9  | 44.9  | 53.8  | 59.5  | 56.6  | 50.7  | 41.8 | 32.8  | 29.9  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 33.3  | 33.7  | 36.0  | 43.6  | 56.1  | 64.2  | 68.3  | 66.7  | 58.1  | 49.1 | 41.1  | 34.9  | (27) |
| (28) |                                                                                  |             | MDCB   | 33.9  | 34.7  | 39.3  | 50.9  | 68.1  | 76.9  | 77.5  | 77.0  | 63.6  | 50.2 | 41.4  | 35.8  | (28) |
| (29) |                                                                                  | 2%          | WB     | 31.5  | 31.8  | 33.1  | 39.5  | 51.6  | 60.3  | 65.5  | 62.3  | 55.3  | 46.1 | 37.0  | 33.2  | (29) |
| (30) |                                                                                  |             | MDCB   | 32.0  | 32.5  | 34.9  | 44.1  | 59.8  | 69.1  | 74.5  | 70.6  | 59.7  | 47.3 | 37.4  | 33.9  | (30) |
| (31) |                                                                                  | 5%          | WB     | 28.8  | 29.4  | 31.4  | 37.2  | 48.5  | 57.1  | 63.3  | 59.8  | 53.1  | 43.9 | 34.7  | 31.8  | (31) |
| (32) |                                                                                  |             | MDCB   | 29.3  | 30.3  | 32.8  | 40.7  | 56.0  | 67.1  | 71.7  | 66.6  | 56.7  | 44.9 | 35.1  | 32.6  | (32) |
| (33) |                                                                                  | 10%         | WB     | 26.3  | 26.6  | 29.5  | 35.4  | 45.5  | 54.6  | 61.1  | 57.5  | 50.8  | 41.7 | 33.0  | 29.9  | (33) |
| (34) |                                                                                  |             | MDCB   | 26.9  | 27.5  | 31.1  | 38.3  | 51.8  | 63.4  | 68.6  | 62.9  | 53.5  | 42.9 | 33.4  | 30.5  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 13.7  | 13.8  | 16.2  | 14.6  | 13.5  | 15.8  | 16.5  | 14.1  | 11.9  | 7.9  | 9.8   | 11.8  | (35) |
| (36) |                                                                                  |             | MCDBR  | 9.1   | 9.7   | 11.8  | 16.2  | 22.9  | 23.2  | 23.4  | 21.8  | 18.0  | 8.6  | 6.4   | 7.9   | (36) |
| (37) |                                                                                  |             | MCWBR  | 9.0   | 9.3   | 9.9   | 11.1  | 13.3  | 11.3  | 11.0  | 11.0  | 11.8  | 7.3  | 6.4   | 7.8   | (37) |
| (38) |                                                                                  | 5% WB       | MCDBR  | 9.1   | 9.9   | 10.2  | 14.2  | 21.2  | 20.3  | 19.8  | 18.5  | 15.9  | 8.0  | 6.2   | 7.9   | (38) |
| (39) |                                                                                  |             | MCWBR  | 9.0   | 9.6   | 9.0   | 10.1  | 12.9  | 11.2  | 10.9  | 10.4  | 11.2  | 7.2  | 6.4   | 7.8   | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.206  | 0.261 | 0.298 | 0.326 | 0.339 | 0.337 | 0.361 | 0.339 | 0.315 | 0.281 | N/A  | N/A   | (40)  |      |
| (41) |                                                                                  | taud        | 1.910  | 2.056 | 2.109 | 2.152 | 2.352 | 2.484 | 2.432 | 2.469 | 2.506 | 2.271 | N/A  | N/A   | (41)  |      |
| (42) |                                                                                  | Ebn at Noon | 41     | 172   | 232   | 259   | 272   | 276   | 263   | 258   | 237   | 169   | N/A  | 0     | (42)  |      |
| (43) |                                                                                  | Edh at Noon | 4      | 16    | 28    | 35    | 32    | 29    | 30    | 26    | 19    | 13    | N/A  | 0     | (43)  |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 16     | 162   | 626   | 1201  | 1452  | 1585  | 1500  | 1015  | 554   | 175   | 33   | 0     | (44)  |      |
| (45) |                                                                                  | RadStd      | 1      | 19    | 38    | 80    | 132   | 169   | 187   | 97    | 57    | 32    | 4    | 0     | (45)  |      |

**Historical Trends**

|                             | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (46) Station Only           | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (47) Regional (6 neighbors) | +1.13 | +4.81   | +4.50  | N/A     | N/A   | N/A   | -232        | -331  | N/A   | N/A   |
|                             | +0.95 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page